Problem Statement

Context / Background: The proliferation of consumer Internet of Things (IoT) devices has scaled rapidly, connecting millions of unpatched, low-power endpoints to residential routers. These compromised devices are frequently hijacked into centralized Command and Control (C&C) botnets to launch large-scale distributed denial-of-service (DDoS) attacks and intercept local network traffic. These botnets operate by establishing communication channels between the infected devices and a central server, which leaves distinct footprints in the network traffic (such as inter-arrival times, flow durations, packet length, and TCP flag counts). AI and machine learning models are now actively used to identify these botnets by analyzing these statistical traffic features, allowing systems to detect malicious behavioral patterns dynamically.

The Problem: While semi-supervised machine learning models offer a theoretical solution for detecting zero-day botnets, their real-world efficiency is severely limited by resource constraints and critically high false-positive rates.

Proposed Solution & Work Plan: To address this gap, this research will evaluate the modern CICIoT2023 dataset, which clearly predefines normal traffic and explicitly labels modern botnet classes such as Mirai, to explore methods including Graph Neural Networks (GNNs), cost-sensitive ensemble learning, and hybrid IDS mechanisms. The objective is to investigate how these approaches define and separate normal versus attack traffic to decrease the false-positive rate, and critically assess whether they can be optimized to maintain the computational efficiency required for lightweight IoT environments.

To achieve this, the research will follow a structured approach:

Project Phase	Approach / Tasks	Interim Report Deliverable
1. Dataset Analysis & Processing	Analyze the CICIoT2023 dataset and filter extraneous telemetry to prevent computational resource exhaustion during the training process.	A finalized, preprocessed dataset ready for machine learning ingestion.
2. Pipeline Verification (Subset Training)	Train and validate the proposed architectures on a small, stratified subset of the dataset to verify code viability and feature selection.	A functional ML pipeline with initial proof-of-concept metrics, ensuring code runs without memory failure.

3. Model Exploration (Scaled)	Build and train the proposed models (e.g., GNNs, Ensembles, hybrid IDS mechanisms) on the larger dataset using the verified parameters.	Baseline detection metrics (Precision, Recall) and initial false-positive rates for the selected models.
4. Efficiency Evaluation	Assess the computational overhead (memory, training time) of the models against the resource constraints of IoT environments.	A completed interim report detailing the trade-off between detection accuracy and computational cost.